4.6 Analyzing Lines of Fit

Name _____ Period ____ Date _____

Algebra I 4.6 · Student sheet

I can…

- use residuals to tell whether a line fits data well.
- use technology to find a line of best fit and the correlation coefficient.
- tell the difference between correlation and causation.

Vocabulary · fill in the missing word

Residual — the _____ between an actual y-value and the y-value predicted by the model (actual − predicted)

Residual plot — a _____ plot of the residuals; a good fit has residuals scattered randomly around 0

Line of best fit — the line found by linear _____

Correlation coefficient r — a number from −1 to 1 that tells how _____ the data fit a line

Causation — a change in one _____ causes a change in another

Part 1 · Check the fit

Line: y = 2.3x + 1.9

x	1	2	3	4	5
Actual y	4	7	8	12	13
Predicted y					
Residual					

Residual plot:

Good fit? _____ Desmos line of best fit: _____ r = _____

Part 2 · Notes and examples (fill in each note, then work the example)

Residual: Residual = _____ − _____.

Example 1

Use y = 2.3x + 1.9 and the data x = 1, 2, 3, 4, 5; y = 4, 7, 8, 12, 13. Find the residuals.

A good fit: A good fit has residuals scattered _____ above and below 0.

Example 2

The residuals from Example 1 are −0.2, 0.5, −0.8, 0.9, and −0.4. Is the line a good fit?

Correlation coefficient: r near 1: _____; r near −1: _____; r near 0: _____.

Example 3

Desmos gives r ≈ 0.98 for the data. Describe the correlation.

Example 4

Cities with more ice cream sales have more sunburns. Does eating ice cream cause sunburns?

Part 3 · Partner practice

Partner 1

A model predicts 50, and the actual value is 54. Find the residual.

Partner 2

Which r shows the strongest relationship: 0.45, −0.91, or 0.80?